

Linear models



1

a $0.5x + 1.5y = 5$

b 4 sheets

2

a \$63.50

b \$5.50

c A fixed cost such as a toll charge.

3

a

Time	0	1	2	3	4
Vertical velocity	120	88	56	24	-8

b -32

c The baseball is moving downwards instead of upwards at $T = 4$.

4

a $y = 45x + 150$

b 45

c The total amount charged increases by \$45 for each additional hour of work.

d 150

e The callout fee or the minimum amount charged by the carpenter.

f \$420

5

a $y = 75x + 50$

b 75

c Mohamad is travelling a distance of 75 mi every hour.

d $y = 50$

e The distance Mohamad had previously driven in his Subaru.

f 218.75 mi

6

a $y = 6.8x + 0.8$

b 6.8

c The cost of each additional minute.

d 0.8

e The initial connection fee

f \$41.60



7

- a $y = 100 - 9x$
- b -9
- c Mario is running at a speed of 9 km/h.
- d 100
- e The length of the ultramarathon or the total distance Mario has to run.
- f 59.5 km

8

- a $y = 16x - 1300$
- b 16
- c The average amount the restaurant receives from each customer.
- d -1300
- e The fixed weekly cost
- f \$3660

9

- a $y = 10x + 340$
- b 10
- c His earnings per phone.
- d 340
- e The minimum amount earned in a week, or the fixed earnings per week independent of how many cellphones are sold.
- f \$700

10

- a 0.2 seconds
- b $y = -0.2x + 6$
- c 5.1 seconds
- d $x = 2.5$ hours

11

- a 1.4 yd/min
- b $y = 1.4x$
- c 8.4 yd
- d 9 minutes

12



a

Number of minutes passed (x)	0	2	4	6	8	10	12
Heart rate (y)	49	55	61	67	73	79	85

b 3

c $y = 3x + 49$

d The change in one minute of Mae's heartbeat.

13

a -0.51 b $y = -0.51x + 40$

c The cost of each text message.

d 40

e The amount of credit Paul purchased.

f \$21.64

14

a

Number of minutes passed (x)	0	5	10	15	20
Amount of fuel left in the tank (y)	150	125	100	75	50

b $y = 150 - 5x$

c The amount of fuel in the car is decreasing at a constant rate of 5 L per minute.

d 30 minutes

15

a $y = 25x + 85$

b 25

c The cost of manufacturing each chair.

d 85

e The fixed cost of manufacturing.

f \$410

16

a $y = -0.6x + 28$ b -0.6

c The rate at which the water level is falling per minute.

d 28

e The water level before any water is pumped out.

f 19 m

Graphs of linear models



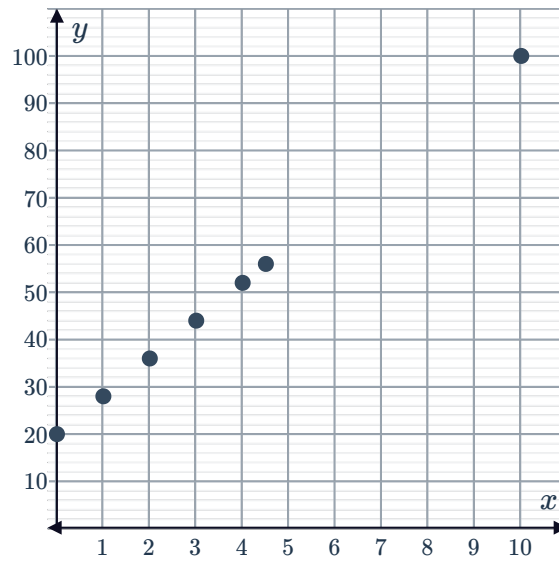
17

a

Number of hours passed (x)	0	1	2	3	4	4.5	10
Amount of water in tank (y)	20	28	36	44	52	56	100

b $y = 8x + 20$

c



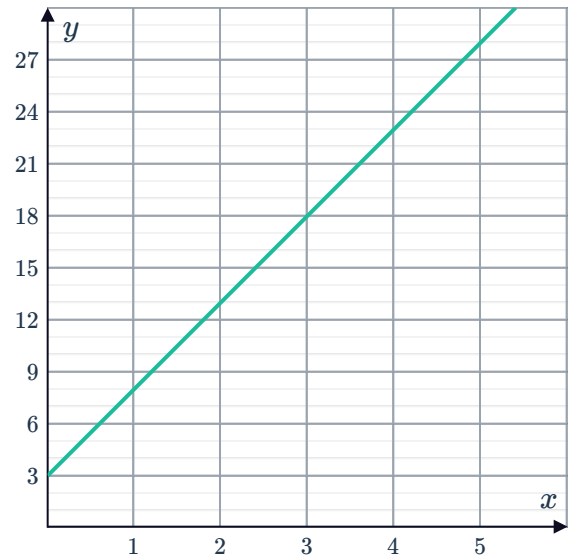
18

a 5°F

c $y = 5x + 3$

b 3°F

d



19

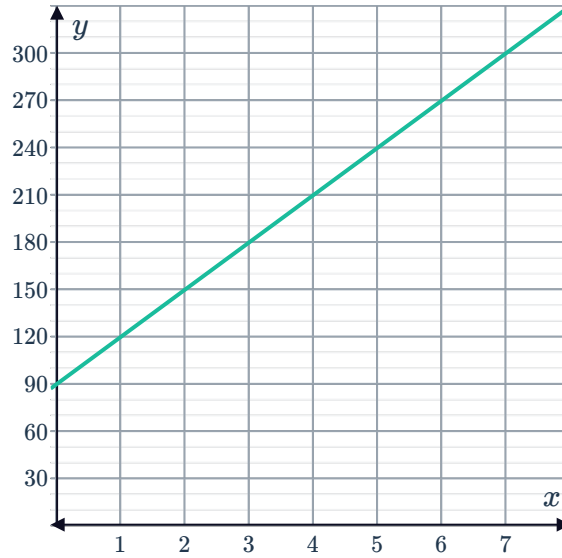


a $y = 30x + 90$

b

Number of hours worked (x)	0	1	2	3	4	4.5	7
Total fee charged (y)	90	120	150	180	210	225	300

c



20

a $\frac{1}{2}$

b 32

c $y = \frac{1}{2}t + 32$

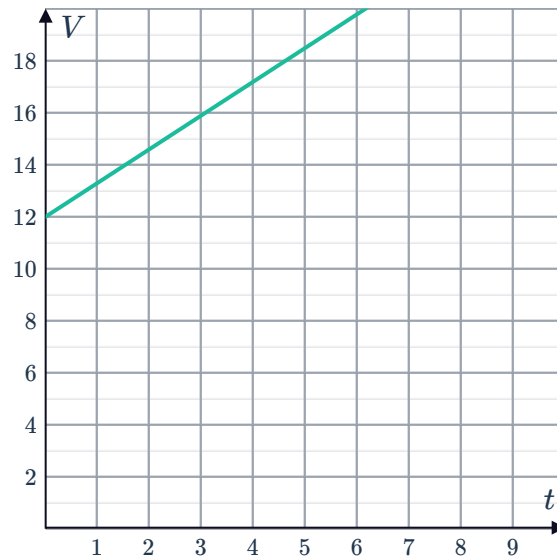
d The amount by which the temperature is rising every minute.

e The temperature of the room before the heater has been turned on.

f 52°F

21

a



b $\frac{13}{10}$

c The rate of acceleration - that is, the rate of increase in the velocity.

d 12

e The initial speed of the ball.

f $V = 12 + \frac{13}{10}t$

g 36.7 m/s

22

a

Water temperature (°F)	-3	-2	-1	0	1	2	3
Surface air temperature (°F)	-7	-5	-3	-1	1	3	5

b $y = 2x - 1$

c 27°F

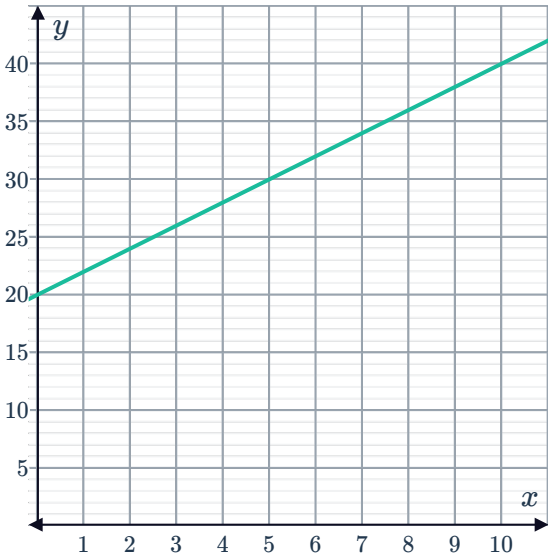
d 12°F



a

Number of hours passed since midnight (x)	0	1	2	3	4	4.5	10
Amount of water in tank (y)	20	22	24	26	28	29	40

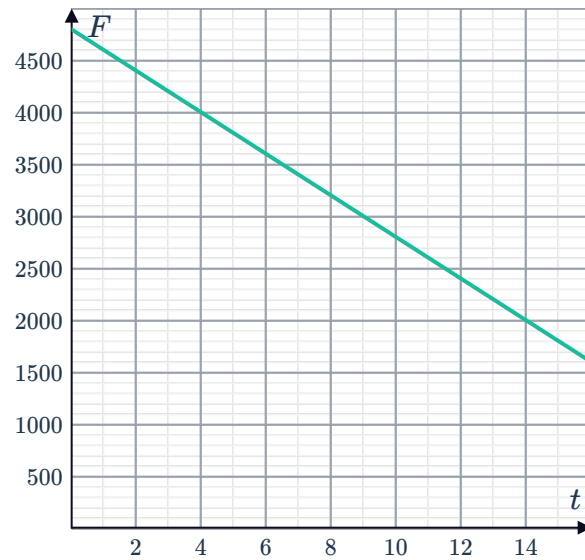
b



- c $y = 2x + 20$
- d 20
- e 2 pm

24

a



b -200

c The rate of change of the fish population each year.

d $F = 4800$

e $F = -200t + 4800$

f 2200 fish

g $t = 14$

25

a $-\frac{1}{2}$

b 30

c $y = -\frac{1}{2}x + 30$

d The amount of water that is flowing out of the hole every minute.

e The capacity of the bucket.

f 3 L